A Complete Characterization of the Strong-Diameter Property for Generalized Limits

Candidate full solution to the concluding question of arXiv:2505.23263

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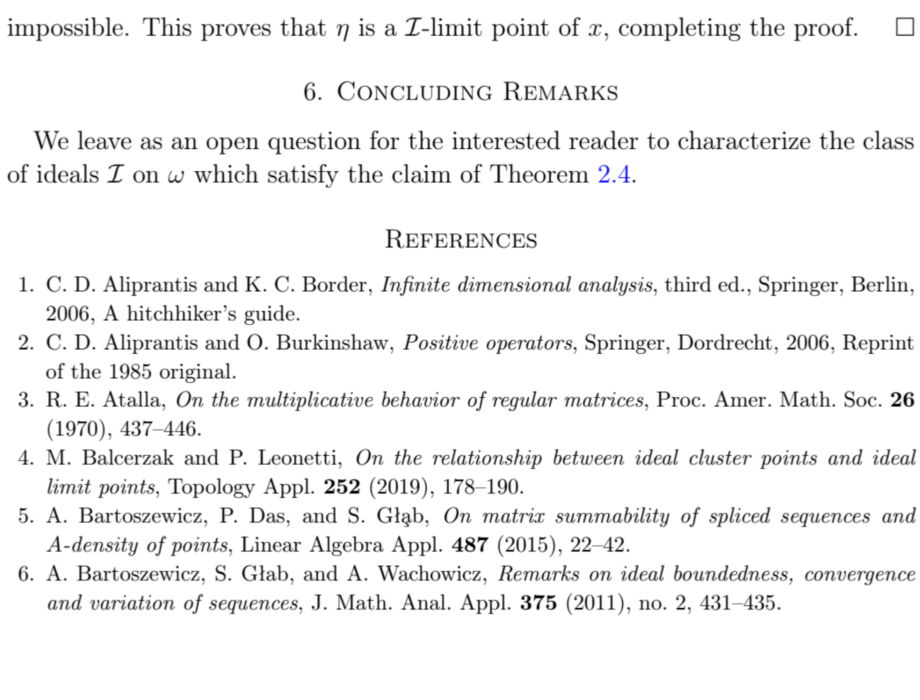
Status. Candidate full solution, likely valid, subject to human review. The result characterizes exactly when every generalized limit has another generalized limit at norm distance two.

1 Source and open question

Leonetti and Orhan study, for an ideal $\mathcal{I}$ on ω containing the finite sets, the family $\mathrm{SL}(\mathcal{I})$ of positive normalized functionals on ℓ_∞ which vanish on $\mathbf{1}_A$ for every $A \in \mathcal{I}$. Their Theorem 2.4 proves that if $\mathcal{I}$ is meager, then

$$\forall f \in \mathrm{SL}(\mathcal{I})\ \exists g \in \mathrm{SL}(\mathcal{I}) \quad \|f - g\| = 2. \tag{1}$$

In Section 6 they ask for a characterization of all ideals satisfying this conclusion [1]. The open question appears on PDF page 14:


impossible. This proves that η is a $\mathcal{I}$-limit point of x, completing the proof. $\square$

6. Concluding Remarks

We leave as an open question for the interested reader to characterize the class of ideals $\mathcal{I}$ on ω which satisfy the claim of Theorem 2.4.

References

1. C. D. Aliprantis and K. C. Border, *Infinite dimensional analysis*, third ed., Springer, Berlin, 2006, A hitchhiker's guide.
2. C. D. Aliprantis and O. Burkinshaw, *Positive operators*, Springer, Dordrecht, 2006, Reprint of the 1985 original.
3. R. E. Atalla, *On the multiplicative behavior of regular matrices*, Proc. Amer. Math. Soc. **26** (1970), 437–446.
4. M. Balcerzak and P. Leonetti, *On the relationship between ideal cluster points and ideal limit points*, Topology Appl. **252** (2019), 178–190.
5. A. Bartoszewicz, P. Das, and S. Głąb, *On matrix summability of spliced sequences and A-density of points*, Linear Algebra Appl. **487** (2015), 22–42.
6. A. Bartoszewicz, S. Głab, and A. Wachowicz, *Remarks on ideal boundedness, convergence and variation of sequences*, J. Math. Anal. Appl. **375** (2011), no. 2, 431–435.

Figure 1: The concluding open question in [1, Section 6, p. 14].

Transcription. "We leave as an open question for the interested reader to characterize the class of ideals $\mathcal{I}$ on ω which satisfy the claim of Theorem 2.4."

2 Answer

For an ultrafilter u on ω, let

$$\mathcal{J}_u = \{A \subseteq \omega : \omega \setminus A \in u\}$$

be its maximal ideal. Following the source, put

$$K = \mathrm{Ult}(\mathcal{I}) = \{u \in \beta\omega : \mathcal{I} \subseteq \mathcal{J}_u\}.$$

Because $\mathcal{I}$ contains the finite sets, $K \subseteq \omega^* = \beta\omega \setminus \omega$.

Theorem 1 (Complete characterization)**.** *For an ideal $\mathcal{I}$ on ω containing* Fin*, the following are equivalent.*

1. *Every $f \in \mathrm{SL}(\mathcal{I})$ has some $g \in \mathrm{SL}(\mathcal{I})$ with $\|f - g\| = 2$.*
2. $\mathrm{Ult}(\mathcal{I})$ *is infinite.*
3. *$\mathcal{I}$ has infinitely many maximal-ideal extensions.*
4. *$\mathcal{I}$ is not the intersection of finitely many maximal ideals.*

Consequently, these conditions give the requested characterization. In particular every meager ideal has the property, as in the source theorem, but meagerness is not necessary.

3 Proof intuition

The compact space $K = \mathrm{Ult}(\mathcal{I})$ is the actual state space behind the problem: Riesz representation identifies $\mathrm{SL}(\mathcal{I})$ with the Radon probability measures on K, and ultrafilter-limit functionals are its Dirac measures.

If K is infinite, it cannot be countable. Indeed, a countably infinite compact Hausdorff space contains a nontrivial convergent sequence, while $\beta\omega$ contains none. A probability measure has only countably many atoms, so an uncountable K contains a point u of measure zero. The Dirac measure at u is then mutually singular to the original measure, hence is at norm distance two.

If K has $m < \infty$ points, take the uniform measure. It assigns positive mass to every point of K, so no probability measure on K is mutually singular to it; quantitatively, its distance from any such measure is at most $2(1 - 1/m) < 2$. This is the exact obstruction.

4 Preparatory lemmas

We include the topological details so that no sequential property of $\beta\omega$ is being used as a black box.

Lemma 2. *$\beta\omega$ is extremally disconnected: the closure of every open set is open.*

Proof. Let $U \subseteq \beta\omega$ be open and set $A = U \cap \omega$. Since ω is dense in $\beta\omega$, A is dense in U, and therefore $\overline{U} = \overline{A}$. For every $A \subseteq \omega$, its closure $\overline{A}$ in $\beta\omega$ is clopen: under the ultrafilter model it is exactly $\{u : A \in u\}$. Hence $\overline{U}$ is open. $\square$

Lemma 3. *$\beta\omega$ contains no nontrivial convergent sequence.*

Proof. Suppose that distinct points x_n converge to x. Passing to a subsequence and using compact Hausdorff regularity, construct decreasing neighborhoods V_n of x and pairwise disjoint open sets U_n such that

$$x_n \in U_n \subseteq V_{n-1} \setminus \overline{V_n}.$$

For completeness, after V_{n-1} has been chosen, take a sufficiently late term $x_n \in V_{n-1}$, choose a neighborhood V_n of x whose closure lies in V_{n-1} and misses x_n, and then choose U_n around x_n inside $V_{n-1} \setminus \overline{V_n}$.

Let $U = \bigcup_n U_{2n}$ and $W = \bigcup_n U_{2n+1}$. They are disjoint open sets. By Lemma 2, $\overline{U}$ is open as well as closed. Since $W \cap \overline{U} = \varnothing$, it follows that $\overline{W} \cap \overline{U} = \varnothing$. But both the even and odd subsequences converge to x, so x belongs to both closures, a contradiction. □

Lemma 4. *Every countably infinite compact Hausdorff space contains a nontrivial convergent sequence.*

Proof. Let X be such a space. It has a nonisolated point x, since otherwise its singleton open cover would have no finite subcover. Enumerate $X \setminus \{x\} = \{y_1, y_2, \ldots\}$. By regularity choose, for each n, a neighborhood O_n of x with $y_n \notin \overline{O_n}$, and set $V_n = \bigcap_{j \le n} O_j$. Then (V_n) is a decreasing local base at x: if O is a neighborhood of x, the compact set $X \setminus O$ is covered by the increasing open sets $X \setminus \overline{V_n}$, hence is covered by one of them.

Every V_n is infinite, for a finite neighborhood of x would make x isolated in a Hausdorff space. Choose distinct $z_n \in V_n \setminus \{x\}$. The local-base property gives $z_n \to x$. □

Corollary 5. *Every infinite closed subspace of $\beta\omega$ is uncountable.*

Proof. A closed subspace is compact. If it were countably infinite, Lemma 4 would give a nontrivial convergent sequence in $\beta\omega$, contradicting Lemma 3. □

5 Proof of the characterization

Lemma 6 (Measure model)**.** *Restriction of the Riesz representation identifies* $\mathrm{SL}(\mathcal{I})$ *affinely and isometrically with the Radon probability measures $\mathcal{P}(K)$ on $K = \mathrm{Ult}(\mathcal{I})$. Under this identification the ultrafilter-limit functional f_u is the Dirac measure δ_u.*

Proof. Use the canonical isometry $\ell_\infty = C(\beta\omega)$. A positive normalized functional f has a unique representing Radon probability measure μ on $\beta\omega$. If $p \notin K$, some $A \in \mathcal{I}$ belongs to p. The clopen neighborhood $\overline{A} = \{u : A \in u\}$ of p has

$$\mu(\overline{A}) = f(\mathbf{1}_A) = 0.$$

Thus every point outside K has a μ-null neighborhood. Inner regularity then gives $\mu(\beta\omega \setminus K) = 0$: every compact subset of this open set has a finite cover by such null neighborhoods. Hence μ is concentrated on K.

Conversely, if $\mu \in \mathcal{P}(K)$ and $A \in \mathcal{I}$, then $\overline{A} \cap K = \varnothing$, so integration against μ annihilates $\mathbf{1}_A$. This yields an element of $\mathrm{SL}(\mathcal{I})$. Riesz representation preserves the dual/total-variation norm, and evaluation at u is represented by δ_u. □

Proof of Theorem 1. First suppose that K is infinite. It is closed in $\beta\omega$, hence uncountable by Corollary 5. Fix $f \in \mathrm{SL}(\mathcal{I})$ and let $\mu \in \mathcal{P}(K)$ represent it. The set

$$\{u \in K : \mu(\{u\}) > 0\}$$

is countable: for each positive integer n, only finitely many points can have mass at least $1/n$. Choose $u \in K$ outside this set. Then $\mu(\{u\}) = 0$, so μ and δ_u are mutually singular. Therefore

$$\|f - f_u\| = \|\mu - \delta_u\|_{\mathrm{TV}} = 2.$$

Since $u \in K$, $f_u \in \mathrm{SL}(\mathcal{I})$. This proves (1).

Now suppose $K = \{u_1, \ldots, u_m\}$ is finite. Let

$$f = \frac{1}{m} \sum_{j=1}^{m} f_{u_j} \in \mathrm{SL}(\mathcal{I}).$$

For any $g \in \mathrm{SL}(\mathcal{I})$, Lemma 6 represents g by weights $a_j \geq 0$ with $\sum_j a_j = 1$. Hence

$$\|f - g\| = \sum_{j=1}^{m} \left| \frac{1}{m} - a_j \right| = 2 \left(1 - \sum_{j=1}^{m} \min \left\{ \frac{1}{m}, a_j \right\} \right) \leq 2 \left(1 - \frac{1}{m} \right) < 2.$$

The inequality holds because some $a_j \geq 1/m$. Thus (1) fails. This proves the equivalence of (1) and (2), while (2) and (3) are equivalent by the definition of K.

Finally, every ideal is the intersection of all maximal ideals containing it. If K is finite, this is a finite intersection. Conversely, suppose $\mathcal{I} = \bigcap_{j=1}^{m} \mathcal{J}_j$ for distinct maximal ideals. If a maximal ideal $\mathcal{J} \supseteq \mathcal{I}$ differed from every $\mathcal{J}_j$, choose $A_j \in \mathcal{J}_j \setminus \mathcal{J}$. The complement of a maximal ideal is an ultrafilter, so $\bigcap_j A_j \notin \mathcal{J}$; but $\bigcap_j A_j \in \mathcal{J}_j$ for every j, hence belongs to $\mathcal{I} \subseteq \mathcal{J}$, a contradiction. Therefore the only maximal extensions are the $\mathcal{J}_j$, and K is finite. This proves (2)$\Leftrightarrow$(4). $\square$

6 Verification and edge cases

- If $\mathcal{I}$ is maximal, then $|K| = 1$ and $\mathrm{SL}(\mathcal{I})$ is a singleton; the theorem correctly says the strong-diameter property fails.
- If $\mathcal{I}$ is an intersection of $m \geq 2$ maximal ideals, the ordinary diameter of $\mathrm{SL}(\mathcal{I})$ is still 2 (take two distinct Dirac measures), in agreement with Theorem 2.3 of the source. What fails is the stronger assertion for the uniform interior point.
- If K is infinite, the proof chooses g to be an extreme point of $\mathrm{SL}(\mathcal{I})$, a statement stronger than the question asks.
- The script `code/verify_finite_distance.py` exhausts rational grids of probability simplices for $1 \leq m \leq 8$ and confirms the sharp finite bound $2(1 - 1/m)$. This is only a sanity check; the displayed formula above is the proof.

7 Novelty check, limitations, and review recommendation

A bounded search on 22 July 2026 used the exact Section 6 wording, the title and authors, and combinations of "Ult($\mathcal{I}$)", "SL($\mathcal{I}$)", strong diameter, and finite maximal-ideal extensions. It located the arXiv version and the 2026 JMAA publication, both retaining the question, but no later resolution. This is not an exhaustive novelty claim.

There are no remaining mathematical cases in the stated question. Human review should focus on Lemma 6, the short topological chain Lemmas 2–4, and the final equivalence with finite intersections. All dependencies are proved in the packet.

References

[1] P. Leonetti and C. Orhan, *On generalized limits and ultrafilters*, arXiv:2505.23263 (2025); J. Math. Anal. Appl. **556** (2026), 130090.